SEGi UNIVERSITY

BCC1313 – ALGORITHMS & COMPLEXITY

Tutorial 8

1. What is the primary difference between comparison-based and non-comparison-based sorting algorithms?
2. Explain how Selection Sort determines the next element to place in its sorted portion.
3. Describe how Quick Sort divides the input array during its sorting process.
4. Given the following unsorted array: [45, 22, 11, 88, 33, 55, 19, 60]
   i) Sort the array using Selection Sort and show all the steps involved in the sorting process.
   ii) Sort the array using Bubble Sort and illustrate the steps for each pass.
   iii) Sort the array using Insertion Sort, and show the state of the array after each element is inserted into the sorted portion.
   iv) Sort the array using Quick Sort and show the steps involved, including the partitioning process and the final sorted array.
   v) Sort the array using Merge Sort and show the steps for splitting the array into smaller subarrays and merging them back together in sorted order.